

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA43

COURSE OUTCOME COVERED

CO 3: Implement dynamic web applications by utilizing DOM-based client-side scripting and employing Java Servlets for server-side request processing and response handling.(L3)

Assessment Tool : MCQ TEST

Weightage: 10%

QUESTIONS:

Total Marks: 20

1. A web application receives multiple client requests simultaneously. The developer wants a centralized method to process all HTTP request types efficiently within a servlet, avoiding duplication of logic in doGet() and doPost() methods. Which servlet method should be overridden to achieve this unified request handling mechanism?

- A) init()
- B) destroy()
- C) service()
- D) doPost()

Answer : C) service() - Handles all requests

2. A servlet-based application establishes a database connection only once during initialization to improve performance and avoid repeated overhead. The developer ensures initialization occurs before servicing requests and only once in the lifecycle. Which method is appropriate for this purpose?

- A) service()
- B) doGet()
- C) init()
- D) destroy()

Answer :

C) init()

Runs once during initialization

3. A banking web application transmits sensitive user credentials. The developer must ensure data is not visible in the URL or cached. Which HTTP method should be used in the servlet to securely transmit such sensitive information?

- A) GET
- B) POST
- C) PUT
- D) HEAD

Answer : B) POST - Sends data in request body

4. A search application allows bookmarking of query URLs. The developer wants parameters appended to the URL for reuse and sharing. Which HTTP method is most suitable for this requirement in servlet-based form submission?

- A) POST
- B) GET
- C) DELETE
- D) PUT

Answer : B) GET - Parameters appear in URL

5. An e-commerce website must maintain shopping cart data across pages without storing it on the client. The developer needs server-side persistence during interaction. Which technique is most appropriate for session tracking in this scenario?

- A) Cookies
- B) Hidden fields
- C) HttpSession
- D) URL rewriting

Answer :

6. A portal stores user preferences on the client to reduce server load and persist across sessions. The data should be lightweight and accessible on revisit. Which mechanism best suits this requirement?

- A) HttpSession
- B) Cookies
- C) JDBC
- D) ServletConfig

Answer : C) HttpSession - Server-side session tracking

7. A user logs in and session data disappears after closing the browser. No persistent storage is configured. What type of behavior is observed for this storage mechanism?

- A) Persistent cookie
- B) Session cookie
- C) Secure cookie
- D) HttpOnly cookie

Answer : B) Session cookie - Removed when browser session ends

8. A developer builds a Java web app interacting with a relational database to retrieve records. A standardized API is required for executing SQL and managing connections. Which technology should be used?

- A) Servlet API
- B) JDBC

C) JSTL

D) JSP

Answer : B) JDBC - Java database connectivity

9. A Java program connects to a database and executes SQL queries. The developer needs an object responsible for sending SQL to the database and executing it. Which JDBC component is used?

A) Connection

B) Statement

C) DriverManager

D) ResultSet

Answer : B) Statement - Executes SQL statements

10. After executing a SELECT query, the developer needs to iterate through returned rows. Which JDBC object stores results and allows traversal of records?

A) Statement

B) Connection

C) ResultSet

D) Driver

Answer : C) ResultSet - Holds SELECT query results

11. A JSP page embeds Java code within HTML to generate dynamic content. The developer uses special tags for this embedded logic. What is this embedded Java code called?

A) Directive

B) Scriptlet

C) Expression

D) JSTL

Answer : B) Scriptlet - Java code inside JSP `<% %>`

12. A web app requires reusable headers and footers across JSP pages. The developer wants inclusion without code duplication. Which JSP feature should be used?

A) Scriptlet

B) Include Directive

C) Expression

D) JSTL

Answer : B) Include Directive - Reusable JSP content

13. A developer aims to reduce Java code inside JSP pages for maintainability, preferring tag-based logic and iteration. Which technology best achieves this?

A) Scriptlets

B) JSTL

C) Servlets

D) JDBC

Answer : B) JSTL - Tag-based JSP logic

14. A JSP page needs to iterate through a product list without embedding Java code. Which JSTL tag should be used for iteration?

A) `<c:if>`

B) `<c:forEach>`

C) `<c:set>`

D) `<c:out>`

Answer : B) `<c:forEach>` - Iterates over collections

15. A web application runs on Apache Tomcat handling servlet execution and requests. What primary role does Tomcat play in this architecture?

A) Database server

B) Web container

C) Compiler

D) IDE

Answer : B) Web container - Tomcat runs/manages servlets

16. A servlet instance is being removed during server shutdown. The developer must release resources such as connections. Which lifecycle method performs cleanup?

A) `init()`

B) `service()`

C) `destroy()`

D) `doGet()`

Answer : C) `destroy()` - Cleanup before servlet removal

17. A developer observes `init()` runs once while `service()` runs for multiple requests. What does this indicate about servlet execution?

A) Multiple instances created

B) Single instance handles requests

C) Error in servlet

D) Session failure

Answer : B) Single instance handles requests - Typical servlet lifecycle

18. A user session expires after inactivity and stored data is lost. What happens to session data after expiration?

A) Persisted permanently

B) Deleted automatically

C) Stored in cookie

D) Sent to database

Answer : B) Deleted automatically - Session data lost after expiration

19. Cookies are disabled in a browser, but session tracking is required. The app uses URL rewriting. What is appended to maintain continuity?

A) Cookie ID

B) Session ID

C) Token

D) Header

Answer : B) Session ID - URL rewriting maintains session

20. A Java web app establishes a database connection dynamically. Which method is responsible for creating the connection?

A) `executeQuery()`

B) `getConnection()`

C) close()
D) prepareStatement()
Answer : B) getConnection() - Creates JDBC connection

Code of Conduct:

*I T. Nikhil Kumar Reddy (192372024) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: T. Nikhil Kumar Reddy

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Accuracy of Responses	40	All or nearly all answers correct (90–100%), demonstrates strong conceptual mastery	Majority correct (75–89%), good understanding	About half correct (50–74%), basic understanding	Few correct (<50%), poor understanding
Conceptual Understanding	25	Demonstrates clear understanding of all concepts behind answers	Good understanding with minor gaps	Basic understanding with some misconceptions	Limited or incorrect understanding of concepts
Application Ability	15	Accurately applies concepts to all application-based MCQs	Applies concepts correctly in most cases	Limited ability to apply concepts	Unable to apply concepts correctly

Time Management	10	Completes all questions efficiently within time	Completes most questions on time	Struggles to complete within time	Unable to complete test within time
Consistency & Attempt	10	Attempts all questions with consistent accuracy	Attempts most questions with reasonable accuracy	Attempts some questions; inconsistent performance	Very few attempts; highly inconsistent